
Patterns

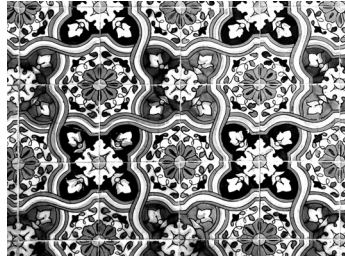
We can make something abstract, then find patterns, then ask if those patterns are caused by some abstract structure. This chapter still has little formality.

3.1 Mathematics as pattern spotting

Patterns are aesthetically pleasing, but they are also about efficiency. They are a way to use a small amount of information to generate a larger amount of information, or a small amount of brain power to understand a large amount of information.

Humans have used patterns across history and across most, if not all, cultures. Patterns are used for designs on fabric, on floors, on walls.

One carefully designed tile can be used to generate a large and intricate pattern covering a very large surface, such as in this tiled wall at the Presidential Palace of Panama. If you look closely you can see that the pattern is actually made from just one tile, rotated and placed at different orientations.[†]



Using one tile like this requires much less “information” than drawing an entire mural from scratch. But aside from this sort of efficiency, patterns can be very satisfying. They give our brains something to latch onto, so that they don’t get too overwhelmed. It’s like when a chorus or refrain comes back in between each verse of a song.

Mathematics often involves spotting patterns. This can help us understand what is going on in general, so that we can use less of our brain power to understand more things.

[†] It might be hard to see in black-and-white, but you can see a color version at this link: eugeniacheng.com/tile/.

Things To Think About

T 3.1 Patterns in the way we write numbers can help us quickly understand some characteristics of individual numbers. What patterns are there in multiples of 10? What about multiples of 5? Multiples of 2? Why are these patterns more noticeable than the patterns in multiples of 3 or 7?

Multiples of 10 all end in 0: 10, 20, 30, 40, and so on. This is something we learn when we are quite small, typically, and it helps us to multiply things by 10 quickly, by sticking 0 on the end. Counting in 10's is something that children are usually encouraged to learn how to do, but if they just do it by "rote" then it might not be as powerful as understanding the pattern, and understanding *why* multiplication by 10 works in this way.

Multiples of 5 end in 5 or 0, and these alternate. Multiples of 2 end in any of the possible even digits: 0, 2, 4, 6, 8. These patterns are much more obvious than the pattern for multiples of 3, which can end in any digit, and go in this order: 3, 6, 9, 2, 5, 8, 1, 4, 7, 0. Likewise multiples of 7, but in a different order.

These patterns don't exactly tell us anything about the numbers 10, 5, 2, 3, and 7 by themselves, but rather, they tell us about those numbers *in the context* of their relationship with the number 10. This is because the way we normally write numbers is based on powers of 10: called base 10, or the decimal number system. This means that the last digit is just 1's (that is, 10 to the power of 0), the second to last digit is 10's, then 100's, then 1000's and so on. So a four-digit number $abcd$ is evaluated as $1000a + 100b + 10c + d$.

The fact that multiples of 10 end in 0 is nothing to do with any inherent property of the number 10, but is because we have chosen to write numbers in base 10. If we wrote numbers in base 9 then it would be all the multiples of 9 that end in 0, and multiplying by 9 is what would be done by sticking a 0 on the end. Similarly the fact that multiples of 5 and 2 have such a tidy pattern in their last digit is nothing to do with any inherent property of 5 and 2, but because of their *relationship* with 10: they are factors.

Technicalities

In base 9, the last digit would still indicate the 1's but then the second to last digit would be 9's, and the third to last would be 81's, and then 729's, and so on. If we write this down in general, then a four-digit base 9 number $abcd$ would "translate" back into an ordinary base 10 number as: $729a + 81b + 9c + d$.

The multiples of 3 in base 9 look like: 0, 3, 6, 10, 13, 16, 20, 23, 26, ... The last digit has a repeating pattern because of how 3 is related to 9 (it is a factor).

Things To Think About

T 3.2 Here is a table for how we add up hours on a 12-hour clock. 1 hour past 12 o'clock is 1 o'clock. 4 hours past 10 o'clock is 2 o'clock. See if you can fill in the rest of the table and then observe some patterns. What would happen on a 24-hour clock? It would take a long time to fill in and would quickly become repetitive, so can you describe the principles governing it instead?

	1	2	3	4	5	6	7	8	9	10	11	12
1	2	3	4	5	6	7	8	9	10	11	12	1
2												
3												
4										2		
5												
6												
7												
8												
9												
10												
11												
12												

For the patterns on the 12-hour clock, here is the rest of the table filled in. The numbers form diagonal stripes, with the whole row of numbers “shifting” over to the left as we move down the table, and the number that falls off on the left-hand side re-appears at the right in the next row down.

	1	2	3	4	5	6	7	8	9	10	11	12
1	2	3	4	5	6	7	8	9	10	11	12	1
2	3	4	5	6	7	8	9	10	11	12	1	2
3	4	5	6	7	8	9	10	11	12	1	2	3
4	5	6	7	8	9	10	11	12	1	2	3	4
5	6	7	8	9	10	11	12	1	2	3	4	5
6	7	8	9	10	11	12	1	2	3	4	5	6
7	8	9	10	11	12	1	2	3	4	5	6	7
8	9	10	11	12	1	2	3	4	5	6	7	8
9	10	11	12	1	2	3	4	5	6	7	8	9
10	11	12	1	2	3	4	5	6	7	8	9	10
11	12	1	2	3	4	5	6	7	8	9	10	11
12	1	2	3	4	5	6	7	8	9	10	11	12

This will be similar on the 24-hour clock so we really don't need to draw it all out. In fact, although we don't usually consider other numbers of hours per day, we could *imagine* clocks with other numbers of hours, say, a 10-hour clock, or a 4-hour clock, or indeed an n -hour clock for any whole number n . Regardless of how many hours there are, the pattern would still be in some sense “the same” on the table, with those diagonals made by numbers shifting over to the left row by row. That principle is quite a deep mathematical structure. We will come back to exactly what the structure is, but for now I want to talk more about the idea of what patterns are and what they tell us.

3.2 Patterns as analogies

Patterns are really analogies between different situations, which is what category theory is going to be all about. At a visual level, we could talk about “stripes”, for example, and understand that we mean alternating lines in different colors. We might not know the specific details – how wide the lines are, what the colors are, what direction they’re pointing – but there is something analogous going on between all different situations involving stripes, and the abstract concept behind it is the concept of a “stripe”. Patterns for clothes are also analogies, this time between different items of clothing. A dress pattern is like an abstract version of the dress.

In our previous examples of numbers we were looking at analogies between numbers. With the repeating patterns of final digits on multiples of numbers it was analogies between multiples. The numbers 10, 20, 30, 40, 50, and so on are all analogous via the fact that they consist of two digits: some number n followed by 0.

With the 12-hour clock the pattern on the table was an analogy between the different rows: in each row, the numbers go up one by one to the right (and the number 1 counts as “one up” from the number 12). The only difference between the rows is then what number they start at.

The analogy between clocks with different numbers of hours is then a sort of meta-pattern — an analogy between different patterns. The thing the different tables have in common is the “shifting diagonal” pattern. We have, in a sense, gone up another level of abstraction to spot a pattern among patterns.[†]

Things To Think About

T 3.3 What patterns can you think of in other parts of life? In what sense could they be thought of as analogies between different situations? You could think about patterns in music, social behavior, politics, history, virus spread, language (in terms of vocabulary and grammar). . .

3.3 Patterns as signs of structure

Spotting patterns in math is often a starting point for developing a theory. We take the pattern as a sign of some sort of abstract structure, and we investigate what abstract structure is causing that pattern.

[†] At this point in writing I went into lockdown for the COVID-19 crisis and finished the draft without leaving the house again. I feel the need to mark it here.

Things To Think About

T 3.4 Here is an addition table for the numbers 1 to 4. Can you find a line of symmetry on this table, that is, a line where we could fold the grid in half and the two sides would match up. Why is that line of symmetry there? What about in a multiplication table for the same numbers?

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Here's the addition table for the numbers 1 to 4 with a line of symmetry marked. I have also highlighted an example of a pair of numbers that correspond to each other according to this line of reflection. The one on the lower left is the entry for $3 + 1$ whereas the one on the upper right is the entry for $1 + 3$.

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

We can *see* that these entries are the same, but the *reason* they are the same is that $3 + 1 = 1 + 3$, which we might know as the commutativity of addition. If you check any other pair of numbers that correspond under the symmetry, you will find that they are all pairs of the form $a + b$ and $b + a$. The entries on the diagonal where the line is actually drawn are all examples of $x + x$ so switching the order doesn't change the entry. That is a form of symmetry in itself: in the expression $a + b = b + a$, if a and b are both x then the left and right become the same. We say the equation is symmetric in a and b .

An analogous phenomenon happens in the multiplication table, with the line of symmetry now being a visual sign of commutativity of multiplication. Often when we spot visual patterns we ask ourselves what abstract or algebraic structure is giving rise to that visual pattern.

Things To Think About

T 3.5 Here is a grid of the numbers 0 to 99. We have already talked about the patterns for multiples of 2, 5, and 10. If we mark in all the multiples of 3, what pattern arises and why? What about multiples of 9?

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99

Here is a picture of the multiples of 3 on the number grid. When we only listed their last digits it was less obvious how much of a pattern there was, because it seemed a bit random: 0, 3, 6, 9, 2, 5, 8, 1, 4, 7. However, when we draw them on this square it's visually quite striking that the multiples of 3 go in diagonal stripes, a bit like the diagonal stripe pattern we saw on the addition table above.

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99

This is because the last multiple of 3 under 10 is one less than it, and so when we move down a row in this table, the pattern shifts one to the left. If the last multiple were 2 less than it, then the pattern would shift two to the left and be less striking. In this picture of multiples of 9 we see something similar happening, just with fewer stripes.

This pattern is essentially where we get that cute trick for the 9 times table where you hold down one finger at a time and read the number off the remaining fingers. So for 4×9 you can hold up all 10 fingers, then put down the fourth one from the left, and read off 3 from the left of that and 6 from the right to get 36. Tricks like that can be a way to bypass understanding or a way to deepen understanding.

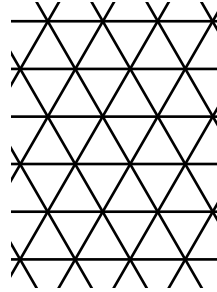
0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99

Things To Think About

T 3.6 How does that trick generalize for other multiples? We will have to change what base we're working in.

A general principle of pattern spotting is that *visual* patterns might be easy to spot in simple examples, but the *abstract* structure might be easier to reason with, use or even verify, in more complex situations. It would be harder to draw the table for multiples of a much larger number, and if the patterns in the table were less obvious then it would be harder to see them. In situations of higher dimensions it is then even harder.

It is fairly easy to see all sorts of patterns of different sized triangles in this grid. However if this were a 3-dimensional space filled with triangular pyramids it would be rather harder to see, and in higher-dimensional space we can't even fit it into our physical world. But those are very helpful structures in many fields of research; it just requires more abstract ways of expressing them.



3.4 Abstract structure as a type of pattern

If category theory is the mathematics of mathematics, then categorical structure is about patterns in patterns. It's about seeing the same pattern in different places, and about making analogies between patterns.

Here's an example. We might talk about a "mother–daughter" relationship abstractly, as opposed to thinking about a specific mother's relationship with their daughter. Now we could think about a relationship between someone's grandmother and mother. This is another type of "mother–daughter" relationship; it's just a particular type where another generation also does exist. The difference this makes to our considerations depends on the context.

Here's a tiny family tree for that structure. In a family tree we're not taking into account any context other than parent–child relationships. In the diagram there is no difference in the abstract structure depicted between the grandmother and her daughter, or the mother and her daughter.

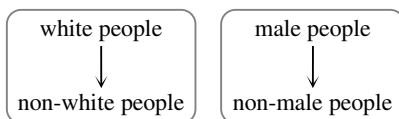


However, if we're writing a book about sociology, or the psychology of family units, or about motherhood, then we might well want to think about how the relationship between a mother and her daughter changes when the daughter has her own daughter in turn. However, the abstract similarity between the grandmother–mother and the mother–daughter relationships is still an aspect of what frames this question.

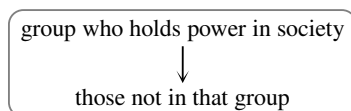
Another example is if we think about power dynamics between different groups of people in society. White people hold structural power over non-white people, and male people hold structural power over non-male people. This is about overall structures, not individuals — it doesn't mean that every white person holds power over every individual non-white person, or that every male person holds power over each individual non-male person. It's about the way the structures of society are set up. In any case even if you dispute this fact

we can still depict the abstract structure that I am describing, because we can describe abstract structure as a separate issue from the question of how the abstract structure manifests itself in “real life”.

We could depict these power structures like this, emphasizing the analogy that I am claiming exists between the two structures.



We could emphasize it further by going one level more abstract to this, which immediately unifies many situations.



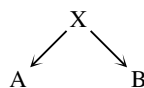
This now includes all sorts of other examples such as straight people over non-straight people, cisgender[†] people over trans people, rich people over non-rich people, educated people over non-educated people, employed over unemployed, people with homes over people experiencing homelessness, and so on.

3.5 Abstraction helps us see patterns

Finding the abstract structure in situations and expressing it in some way, often as a diagram to make it less abstract, can help us see the patterns and relationships between situations. It can help clear our mind of clutter and distraction and emotions. Distraction and emotions aren’t bad per se, they can just get in the way of us seeing the actual structure of an argument rather than the window-dressing. Sleight of hand and flattering clothing can be fun, but if we’re at the doctor’s getting diagnosed for something it would be much more productive to show the whole truth and not be afraid of getting naked.

One of my favorite examples is the way I have been drawing diagrams of analogies and levels of abstraction. This way of specifying their structure has helped me pin down much more clearly where disagreements around analogies are coming from. I described it in *The Art of Logic* in terms of disagreements largely taking two possible forms.

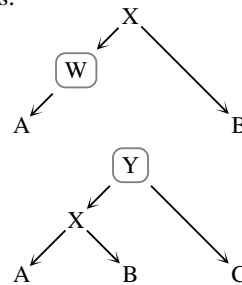
It starts by someone invoking an analogy in this form, between two ideas A and B, but crucially without specifying what abstract level X they’re referring to.



[†] Cisgender people are those whose gender identity matches the one they were assigned at birth.

Someone then objects, for either of these two reasons:

1. They see a more *specific* principle W at work that is really the reason behind A, so they do not consider A and B to be analogous:
2. They see a more *general* principle Y that they think the first person is invoking. This makes some other thing C analogous, and they object to that.



Things To Think About

T 3.7 Try modeling these two arguments with the above abstract structures:

1. Some people say COVID-19 is just like the normal seasonal flu. However medical professionals and scientists (and I) say it is not.
2. Some people (including me) believe that same-sex marriage is just like opposite-sex marriage, but others erroneously think that this means we also accept incest and pedophilia.

For COVID-19, we have the first case: in the diagram A is COVID-19 and B is seasonal flu. Some people believe X, that they are both contagious viruses that cause respiratory problems. This is true. However others recognize W, which is that COVID-19 is a new virus so there is only a new vaccine, no herd immunity, and less understanding of how to treat it (at time of writing). This is aside from it being more contagious and more deadly across the population.

For the second example, A is opposite-sex marriage and B is same-sex marriage. Some of us believe X, that marriage is between any two consenting unrelated adults. But others hallucinate that we believe a further level Y of just “adults” or even “people” which would then include examples C such as incest or pedophilia. But X and Y are very different levels.

These structures are abstract, and the abstraction allows me to see the pattern in common between essentially all arguments in which analogies are disputed. It also allows me to see how to make these arguments better: by being clearer about what the principles at work really are, and to explore the sense in which the different cases are and aren’t analogous, rather than just declaring that something is or is not analogous.

If we aren’t clear about different levels of abstraction then it causes angry disagreement. This is particularly a shame because different levels of abstraction are inevitable and indeed important. Different patterns might be relevant in different situations, and so we really *need* to be able to invoke different abstractions in different contexts. In the next chapter we’ll look more at the sorts of different patterns that can arise depending on context.